SymPy Bug Card

Bug 24: solveset returns 1 for asec(x) = 2*pi outside the principal asec range

Task. Solve the inverse-secant equation over the complex domain:

$$\operatorname{asec}(x) = 2\pi.$$

SymPy's answer.

$$\operatorname{solveset}(\operatorname{asec}(x) = 2\pi, x, \mathbb{C}) = \{1\}.$$

Correct answer.

$$\varnothing.$$

Explanation. Substitution rejects the returned value: $\operatorname{asec}(1) - 2\pi = -2\pi$. Principal inverse secant satisfies $\operatorname{asec}(1) = \arccos(1) = 0$, so applying sec to both sides has introduced an extraneous solution.

Likely root cause. The diagnosis identifies `sympy/solvers/solveset.py`'s generic `.inverse()` path in `_invert_complex`: asec^{-1} is treated as sec without a principal-range check or final candidate verification. This is a new instance of a known failure mode.

Artifact (GitHub). `bug-report/bug-024-solveset-returns-1-for-asec-x-2-pi-outside-the-principal-ase/`